**REPORT AND PRESENTATION**

**Project Title:** NUL Smart Library System
**Course:** Data Structures and Algorithms
**Institution:** National University of Lesotho
**Language Used:** C++
**Group members:**


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## Abstract


The NUL Smart Library System is a console-based C++ application that manages book borrowing, returns, reservations, and catalog searches for a university library.
It applies fundamental data structures and algorithms such as Binary Search Trees (BSTs), Hash Tables, Queues, and Linked Lists to simulate an efficient and organized digital library management system.


## Objectives

- To design a simple library management system using core data structures.
- To efficiently index, search, and retrieve book information.
- To manage borrowing and returning processes.
- To implement reservation and waitlist functionality.
- To maintain a history of all transactions.

## System Features

| Feature | Description | Data Structure Used |
|---|---|---|
| Book Catalog | Stores and organizes books alphabetically. | **BST** |
| Fast Search | Quickly finds books by title. | **Hash Table** |
| Borrow/Return | Allows users to borrow or return books. | **BST + Linked List** |
| Reservations | Keeps track of users waiting for borrowed books. | **Queue** |
| Transaction Log | Records all borrow/return actions. | **Linked List** |

## System Design

**Main Components:**

1. **BST (Binary Search Tree):**
   - Used for indexing and organizing books alphabetically by title.
   - Supports insertion, searching, and traversal.
2. **Hash Table:**
   - Provides quick lookup of books by title using a hash function.
   - Resolves collisions via chaining.
3. **Queue:**
   - Maintains waiting lists for books currently borrowed.
   - Operates on FIFO (First-In-First-Out) principle.
4. **Linked List:**
   - Used to maintain transaction history (borrowed/returned books).

## Implementation Summary

- **Language:** C++
- **Input/Output:** Console-based interface
- **No STL vectors used** (all ADTs implemented manually)
- **Main Classes/Structures:**
  - `BST` → Book indexing
  - `HashTable` → Fast lookups
  - `Queue` → Waitlists
  - `TransactionList` → Transaction history
  - `LibrarySystem` → Controls program flow and menu operations

## Sample Outputs

### Adding Books:

```
Enter book title: Data Structures
Enter author: Mzwaa
Book added successfully.
```

```
==== NUL SMART LIBRARY SYSTEM ====
1. Add Book
2. Borrow Book
3. Return Book
4. Search Book
5. Show All Books
6. Show Transactions
0. Exit
Enter choice: 1
Enter book title: Data Structures
Enter author: Mzwaa
Book added successfully.
```

### Borrowing Book:

```
Enter your User ID: 202321287
Enter book title: Data Structures
Book borrowed successfully.
```

```
==== NUL SMART LIBRARY SYSTEM ====
1. Add Book
2. Borrow Book
3. Return Book
4. Search Book
5. Show All Books
6. Show Transactions
0. Exit
Enter choice: 2
Enter your User ID: 202321287
Enter book title: Data Structures
Book borrowed successfully.
```

### Returning Book:

```
Enter your User ID: 202321287
Enter book title: Data Structures
Book returned successfully.
```

```
==== NUL SMART LIBRARY SYSTEM ====
1. Add Book
2. Borrow Book
3. Return Book
4. Search Book
5. Show All Books
6. Show Transactions
0. Exit
Enter choice: 3
Enter your User ID: 202321287
Enter book title: Data Structures
Book returned successfully.
```

## Search Books:

```
Enter book title to search: Data Structures
Book found - Title: Data Structures | Author: Mzwaa | Available: Yes

Enter book title to search: C++
Book not found.
```

```
==== NUL SMART LIBRARY SYSTEM ====
1. Add Book
2. Borrow Book
3. Return Book
4. Search Book
5. Show All Books
6. Show Transactions
0. Exit
Enter choice: 4
Enter book title to search: Data Structures
Book found - Title: Data Structures | Author: Mzwaa | Available: Yes

==== NUL SMART LIBRARY SYSTEM ====
1. Add Book
2. Borrow Book
3. Return Book
4. Search Book
5. Show All Books
6. Show Transactions
0. Exit
Enter choice: 4
Enter book title to search: C++
Book not found.
```

## Library books:

```
--- Library Books ---
Title: Data Structures | Author: Mzwaa | Available: Yes
```

```
==== NUL SMART LIBRARY SYSTEM ====
1. Add Book
2. Borrow Book
3. Return Book
4. Search Book
5. Show All Books
6. Show Transactions
0. Exit
Enter choice: 5

--- Library Books ---
Title: Data Structures | Author: Mzwaa | Available: Yes
```

## Transaction History:

```
--- Transaction History ---
User: 202321287 | Book: Data Structures | Action: Borrowed
User: 202321287 | Book: Data Structures | Action: Returned
```

```
==== NUL SMART LIBRARY SYSTEM ====
1. Add Book
2. Borrow Book
3. Return Book
4. Search Book
5. Show All Books
6. Show Transactions
0. Exit
Enter choice: 6

--- Transaction History ---
User: 202321287 | Book: Data Structures | Action: Borrowed
User: 202321287 | Book: Data Structures | Action: Returned
```

## Data Structure Justification

| Data Structure | Justification |
|---|---|
| BST | Efficient organization and traversal of books. |
| Hash Table | Fast access by book title (O(1) average). |
| Queue | Fair handling of waiting lists. |
| Linked List | Dynamic, simple storage of transaction logs. |

## Challenges

- Managing multiple data structures together efficiently.
- Avoiding STL containers while maintaining performance.
- Implementing proper memory management for all nodes.

## Conclusion

The NUL Smart Library System successfully demonstrates the application of fundamental data structures to solve real-world problems.
The integration of BSTs, Hash Tables, Queues, and Linked Lists results in a simple yet powerful system capable of handling core library management operations efficiently.

## Future Improvements

- Replace BST with AVL Tree for automatic balancing.
- Add file handling to save and load library data.
- Build a GUI version using C++ graphics.
- Add user authentication for security.